

## Element C

With both our interview with the Nashville Zoo and our research on the problem and possible solutions as seen in Elements A and B we created some basic criteria/guidelines for both the lift and code part of our project.

### **Criteria:**

#### 1. Simplicity

- a. For whatever we create to be effective it has to be able to be operable even when we are not able to work with it. This means the design must be simple enough for someone to learn how it works with only a design manual. This will be tested by asking the zoo handlers if they understand how the design works.
- b. LCD must always display the proper time

#### 2. Mobility

- a. The base must be able to adjust the feeder at a 180-degree angle horizontally
- b. Must be able to move from the shed to the moat of the enclosure

#### 3. Ease of use

- a. Automation must be able to be started and stopped with only one handler using it
- b. A handler must be able to leave it on for the whole day without any mishaps

#### 4. Timing

- a. Must randomly shoot within designated food times given by the zoo
- b. Must keep time even while offline or not in use

#### 5. Location

- a. Must be able to randomize the location of where the food is shot in the enclosure
- b. Must not land in any area deemed unacceptable by the zoo (water, trees, and moat)

**Constraints:**

- 1. All parts must be bought within the \$150 budget
- 2. The prototype is due the first week of April
- 3. Final design due May 16, 2024
- 4. Design should last multiple years
- 5. Arduino and base should be easy to care for and not break/fail under bad conditions
- 6. Arduino should be easy to access
- 7. Must be able to adapt to changes in bear schedule of feeding

The prior criteria and constraints allow us to feel confident in our efforts to create a code that automates both the randomized shooting and the timing of feeding the bears. Through the interview, we were able to deduce that mobility, timing, location, and ease of use were the most important criteria for the bear handlers. The main constraints that limit our design would be the budget and the fact that the code must be easy to change because the bears don't have a set schedule.

The following are our first ideas and thoughts regarding the constraints and criteria for the design:

